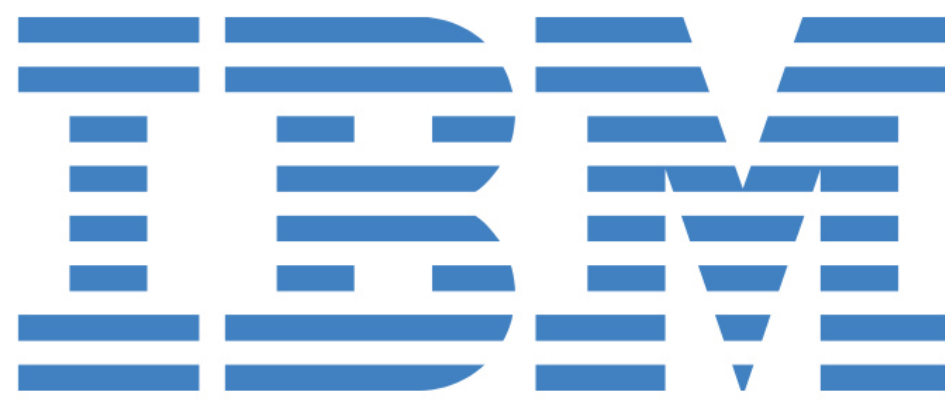


F1 AI-commentary

Telemetry Replay

AI Commentary

Interactive Dashboard



TEAM

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CLIENT

IBM

Client Brief

Turning raw racing telemetry into an interactive replay experience.

In partnership with **IBM**, we developed a platform that makes motorsport telemetry easier to explore, understand, and enjoy. Users can load simulator CSV sessions or selected FastF1 race data, replay the action through an interactive dashboard, inspect key metrics such as RPM, speed, braking, and track position, and follow synchronised AI-generated commentary as each lap unfolds.

Rather than presenting telemetry as static raw numbers, our system transforms it into a clear visual narrative that is engaging for casual viewers while still providing meaningful detail for more technical users.

CORE FEATURE
INTERACTIVE REPLAY

AI LAYER
AI COMMENTARY

USER VALUE
TELEMETRY INSIGHT

Project Snapshot

Flutter

FastAPI

OpenAI

WebSockets

Telemetry

AWS

Dart

Python

Real Data

Core technologies used across frontend, backend, AI integration, and live telemetry playback.

What To Notice

- 01 RPM, speed, steering, brake and fuel shown in one clean telemetry HUD.
- 02 Replay-synchronised AI commentary streamed as live text and audio.
- 03 Simple controls: play, pause, seek, lap select and up to 32x speed.
- 04 Track map with live car position and a speed heatmap overlay.
- 05 Best-lap ghost and time delta for instant performance comparison.
- 06 Two data modes: simulator CSV replay and FastF1 race browsing.

Scan To Explore

Scan to open the website



Or use the link:

dev.f1aicommentary.co.uk

Scan for the detailed docs



Or use the link:

docs.f1aicommentary.co.uk

Note: Scanning the website QR code will let you play with the product, scanning the docs QR code will give you a detailed, behind-the-scenes look at how our project came to be.

Product Preview



Under the Hood

Telemetry is loaded from simulator CSV files or selected FastF1 sessions, replayed through a FastAPI backend, and turned into synchronised visuals and AI commentary inside the Flutter web app.

CSV / FastF1
Telemetry

FastAPI Playback
& WebSockets

OpenAI Realtime
Text + Audio

Flutter Web
Dashboard

- Telemetry can come from parsed simulator CSV sessions or historical FastF1 race data.
- The backend creates replay sessions and streams synchronised telemetry frames over WebSockets, with play, pause, seek, lap jumps and speed control.
- OpenAI Realtime generates short F1-style text and audio commentary, while the Flutter UI renders the HUD, track map, heatmap and lap delta view.

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